

 VOCABULARY GUIDE

**LOOKS SIMILAR.
MEANS DIFFERENT.**

AI Terms That Confuse Everyone

25 AI terms you've been using interchangeably (but
shouldn't)

→ Swipe to fix your vocabulary



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 Repost

Latency **VS** Response Time

🔍 WHAT

Latency: time until the first token/byte arrives.

Response Time: total time until the entire response is complete.

🔄 WHY THE CONFUSION

Both feel like "how fast is it" but one measures a starting point, the other measures a full journey.

🕒 WHEN IT MATTERS

Chat UIs care about latency (feels instant). Batch jobs care about response time (total throughput).

💡 EXAMPLE

A model with 200ms latency but a 10-second response time feels fast (streaming starts quick) even though the full answer takes long.

❌ COMMON MISTAKE

Teams optimize response time and wonder why users still say the app "feels slow" they ignored latency.



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Context Window **vs** Memory

? WHAT

Context Window: the token limit the model can "see" in one session.

Memory: information that persists across sessions.

⌘ WHY THE CONFUSION

Both relate to "what the model remembers" but context window resets, memory doesn't.

🕒 WHEN IT MATTERS

Building a chatbot? Context window decides conversation length. Memory decides if it remembers you tomorrow.

💡 EXAMPLE

ChatGPT's memory feature isn't expanding the context window it's injecting saved facts back into a fresh context window each time.

✖ COMMON MISTAKE

Assuming a bigger context window = the model "remembers" you. It doesn't, once the session ends.



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RAG vs Fine-Tuning

? WHAT

RAG: retrieves external knowledge at query time, no weight changes.

Fine-tuning: retrain the model's weights on new data.

⚡ WHY THE CONFUSION

Both are "ways to make the model know more."

🕒 WHEN IT MATTERS

RAG → fast-changing, factual, document-heavy knowledge.

Fine-tuning → changing tone, behavior, format, or domain reasoning style.

💡 EXAMPLE

A legal chatbot needing today's case law → RAG. A chatbot needing to "sound like a senior partner" → fine-tuning.

✖ COMMON MISTAKE

Fine-tuning a model to "add facts" facts go stale; RAG handles that far better.



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Agent **vs** Workflow

? WHAT

Agent: decides its own next steps using reasoning.

Workflow: follows a fixed, predefined sequence.

⚡ WHY THE CONFUSION

Both "automate a process."

🕒 WHEN IT MATTERS

Predictable, repeatable task → workflow (cheaper, reliable). Open-ended, judgment-based task → agent.

💡 EXAMPLE

"Send invoice every Friday" = workflow. "Investigate why revenue dropped and decide what to check next" = agent.

✖ COMMON MISTAKE

Calling every LLM-powered automation an "agent." Most are just workflows with an LLM step.



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Embedding **VS** Vector

? WHAT

Embedding: the process/technique of converting data into numeric representation.

Vector: the output the actual array of numbers.

⚡ WHY THE CONFUSION

People use them as synonyms in casual speech.

🕒 WHEN IT MATTERS

Talking architecture: "we embed the text" (verb/process) vs "we store the vector" (noun/result).

💡 EXAMPLE

"The embedding model converts text into a 1536-dimension vector" one sentence, both terms, both correct.

✖ COMMON MISTAKE

Saying "vector model" instead of "embedding model" the model embeds; the vector is what it produces.



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MCP **VS** API

🔍 WHAT

API: a general interface contract for two systems to talk.

MCP (Model Context Protocol): a standardized protocol specifically for exposing tools/context to LLMs.

🤖 WHY THE CONFUSION

MCP is technically built on API-like communication, so people think it's "just another API."

🕒 WHEN IT MATTERS

Building one-off integration → API. Building something reusable across multiple AI agents/clients → MCP.

💡 EXAMPLE

A weather API returns JSON for any app. An MCP server exposes that same data in a way any LLM client (Claude, etc.) can discover and use automatically.

❌ COMMON MISTAKE

Thinking MCP replaces APIs. It doesn't MCP often wraps APIs to make them LLM-consumable.



Precision **VS** Recall

? WHAT

Precision: of what you retrieved, how much was relevant.

Recall: of all relevant items that exist, how much did you retrieve.

✂ WHY THE CONFUSION

Both sound like "accuracy" of a retrieval system.

🕒 WHEN IT MATTERS

Search engines often favor precision (top results matter).
Legal/medical search favors recall (can't miss anything).

💡 EXAMPLE

Retrieving 3 docs, all relevant = high precision. But if 10 relevant docs existed, recall is low.

✖ COMMON MISTAKE

Optimizing only for precision in RAG systems leads to confident answers built on incomplete context.



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Tool Calling **VS** Function Calling

? WHAT

Function Calling: OpenAI's specific term/implementation for invoking a single defined function.

Tool Calling: the broader concept functions, APIs, plugins, retrieval, code execution, etc.

✂ WHY THE CONFUSION

Function calling was the first popularized version, so it became the generic term.

🕒 WHEN IT MATTERS

Cross-provider work Claude, Gemini, and others use "tool calling" as the umbrella term; "function calling" is narrower and OpenAI-specific.

💡 EXAMPLE

A "tool" can be a calculator function, a web search, or a database query function calling usually refers to just the function-style invocation.

✖ COMMON MISTAKE

Using the terms interchangeably in technical docs confuses readers across different model providers.



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GPT vs ChatGPT

? WHAT

GPT: the underlying model/architecture (e.g., GPT-4).

ChatGPT: the product the chat interface, memory, plugins, and UX wrapped around the model.

✂ WHY THE CONFUSION

Most people only ever interact with ChatGPT, so they think "GPT" and "ChatGPT" are the same thing.

🕒 WHEN IT MATTERS

Developers use GPT via API (no chat UI, no memory by default). End users use ChatGPT (full product experience).

💡 EXAMPLE

You can use GPT-4 through the API with zero chat history that's not "using ChatGPT."

✖ COMMON MISTAKE

Saying "ChatGPT's API" there is no such thing; it's the GPT model's API.



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Vector Database **vs** Knowledge Graph

? WHAT

Vector Database: stores embeddings, retrieves by similarity.

Knowledge Graph: stores entities and explicit relationships between them.

⌘ WHY THE CONFUSION

Both are "ways to store knowledge for AI to query."

🕒 WHEN IT MATTERS

Fuzzy, semantic similarity search → vector DB. Precise relationship reasoning ("who reports to whom") → knowledge graph.

💡 EXAMPLE

"Find documents similar to this one" → vector DB. "Find all employees who report to someone in Finance" → knowledge graph.

✖ COMMON MISTAKE

Using a vector DB for relationship-heavy queries it can't reliably answer multi-hop relational questions.



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Token **VS** Word

? WHAT

Token: the model's basic unit of text (subword chunks).

Word: a human linguistic unit.

⚡ WHY THE CONFUSION

Often roughly 1:1 for simple English words, so people assume they're equivalent.

🕒 WHEN IT MATTERS

Pricing, context limits, and rate limits are all token-based, not word-based.

💡 EXAMPLE

"Tokenization" might split into "Token" + "ization" one word, two tokens.

✖ COMMON MISTAKE

Estimating costs by word count token count (often ~1.3x word count) is what actually bills you.



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Training **VS** Inference

? WHAT

Training: the process of teaching the model by adjusting weights.

Inference: the process of using the trained model to generate output.

⚡ WHY THE CONFUSION

Both involve "the model doing something with data."

🕒 WHEN IT MATTERS

Training happens once (or periodically); inference happens every single time you send a prompt.

💡 EXAMPLE

GPT-4 was trained over months on massive datasets. Every ChatGPT response you get is inference no learning happens.

✖ COMMON MISTAKE

Thinking the model "learns" from your conversation. It doesn't that's inference, not training.



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Fine-tuning **VS** Prompt Engineering

? WHAT

Fine-tuning: updates the model's internal weights with new examples.

Prompt Engineering: crafts better inputs without touching the model at all.

⚡ WHY THE CONFUSION

Both aim to "get better outputs."

🕒 WHEN IT MATTERS

Need quick iteration, low cost → prompt engineering. Need consistent behavior baked in at scale → fine-tuning.

💡 EXAMPLE

Adding "respond only in bullet points" to your prompt = prompt engineering. Training the model on 10,000 bullet-formatted examples = fine-tuning.

✖ COMMON MISTAKE

Jumping to fine-tuning before exhausting prompt engineering most problems are solvable with better prompts, cheaper and faster.



Similarity Search **VS** Semantic Search

? WHAT

Similarity Search: pure vector-distance matching (cosine similarity, etc.).

Semantic Search: broader system that incorporates meaning/intent, often combining similarity search with ranking, filters, or hybrid methods.

✂ WHY THE CONFUSION

Semantic search is often implemented using similarity search, so the terms blur.

🕒 WHEN IT MATTERS

Designing a search system similarity search is one component; semantic search is the user-facing capability.

💡 EXAMPLE

Pure similarity search might return "Apple the fruit" for a query about "Apple the company" if vectors are close. Semantic search systems add context/ranking to avoid that.

✖ COMMON MISTAKE

Calling raw cosine-similarity retrieval "semantic search" it's a building block, not the whole system.



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Dense Retrieval **VS** Sparse Retrieval

? WHAT

Dense Retrieval: uses embeddings to find conceptually similar text.

Sparse Retrieval: uses keyword/term matching (e.g., BM25, TF-IDF).

⚡ WHY THE CONFUSION

Both are "retrieval methods" used in RAG pipelines.

🕒 WHEN IT MATTERS

Conceptual/paraphrased queries → dense. Exact keyword/legal/code search → sparse. Best systems use both (hybrid retrieval).

💡 EXAMPLE

Searching "car problems" finding a doc about "automobile issues" = dense retrieval working. Searching an exact product SKU = sparse retrieval winning.

❌ COMMON MISTAKE

Going all-in on dense retrieval and losing exact-match accuracy for codes, IDs, or names.



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Reflection **VS** Self-Correction

? WHAT

Reflection: the model evaluates/critiques its own output.

Self-Correction: the model acts on that critique to produce a better output.

⌘ WHY THE CONFUSION

They're sequential steps of the same loop, so people merge them into one idea.

🕒 WHEN IT MATTERS

Designing agent loops reflection without self-correction is just commentary; you need both for actual improvement.

💡 EXAMPLE

"This answer is missing a citation" = reflection. Regenerating the answer with the citation added = self-correction.

✖ COMMON MISTAKE

Building a "reflection step" that never feeds back into the generation wasted compute, no improvement.



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Autonomous **VS** Automated

? WHAT

Automated: follows fixed, predefined rules no decision-making.

Autonomous: makes independent decisions based on context/reasoning.

⌘ WHY THE CONFUSION

Both mean "runs without a human."

🕒 WHEN IT MATTERS

Marketing vs. reality many "autonomous agents" are actually just automated scripts with an LLM bolted on.

💡 EXAMPLE

A cron job emailing a report = automated. An agent that decides whether a report is even needed today = autonomous.

✖ COMMON MISTAKE

Overselling automation as autonomy sets wrong expectations for what the system can actually handle.



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Hallucination VS Bias

? WHAT

Hallucination: the model generates false/fabricated information confidently.

Bias: the model's outputs are systematically skewed toward certain patterns, viewpoints, or groups.

⚡ WHY THE CONFUSION

Both are "model gets it wrong" but one is about truth, the other about fairness/skew.

🕒 WHEN IT MATTERS

Debugging output errors fact-check for hallucination, audit for bias; different fixes for each.

💡 EXAMPLE

Model citing a fake research paper = hallucination. Model consistently associating certain job titles with one gender = bias.

✖ COMMON MISTAKE

Lumping all "wrong" outputs under "hallucination" bias often needs dataset-level fixes, not fact-checking.



Accuracy VS Precision

? WHAT

Accuracy: overall correctness across all predictions (right vs total).

Precision: correctness specifically among positive predictions.

⚡ WHY THE CONFUSION

Both sound like "how often is the model right."

🕒 WHEN IT MATTERS

Imbalanced datasets accuracy can be misleading; precision tells you a sharper story.

💡 EXAMPLE

A spam filter that marks everything "not spam" can have 95% accuracy (if spam is rare) but 0% precision on catching spam.

✖ COMMON MISTAKE

Reporting only accuracy on imbalanced classification tasks it hides real performance issues.



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Monitoring **VS** Observability

? WHAT

Monitoring: tracking known, predefined metrics (latency, error rate, token usage).

Observability: the ability to debug unknown issues using logs, traces, and system internals.

⌘ WHY THE CONFUSION

Both are "watching the system."

🕒 WHEN IT MATTERS

Monitoring tells you something broke. Observability tells you why it broke.

💡 EXAMPLE

Monitoring shows latency spiked at 2pm. Observability lets you trace it to a specific slow tool call inside an agent chain.

✖ COMMON MISTAKE

Having dashboards (monitoring) but no tracing (observability) you see the symptom, never the cause.



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Open Source **VS** Open Weight

? WHAT

Open Source: full transparency code, training data, methodology all available.

Open Weight: only the trained model parameters are released; training data/code stay private.

⚡ WHY THE CONFUSION

Both get marketed as "open."

🕒 WHEN IT MATTERS

Compliance, auditing, and reproducibility open weight models can't be fully audited or retrained from scratch.

💡 EXAMPLE

Llama models are open weight (you get the weights, not the full training pipeline or dataset).

✖ COMMON MISTAKE

Calling open-weight models "open source" a common but technically inaccurate label.



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Quantization **VS** Compression

? WHAT

Quantization: reduces numeric precision of weights (e.g., 32-bit → 4-bit).

Compression: the broader umbrella includes quantization, pruning, distillation, and more.

⚡ WHY THE CONFUSION

Quantization is the most common compression technique, so people use it as a stand-in for the whole category.

🕒 WHEN IT MATTERS

Choosing a model-shrinking strategy quantization is one tool, not the only one.

💡 EXAMPLE

Running a 4-bit quantized model on a laptop = quantization. Combining quantization + pruning + distillation for a mobile model = full compression pipeline.

✖ COMMON MISTAKE

Saying "we compressed the model" when you only quantized it leaves out other available techniques.



Temperature **VS** Top-P

? WHAT

Temperature: scales the entire probability distribution to control randomness.

Top-P (nucleus sampling): truncates the distribution to the smallest set of tokens whose cumulative probability hits P.

✂ WHY THE CONFUSION

Both are "randomness knobs" for generation.

🕒 WHEN IT MATTERS

Tuning creative vs. deterministic output they interact, and tuning both without understanding each can give unpredictable results.

💡 EXAMPLE

Temperature 0.9 + Top-P 1.0 = broad, creative sampling. Temperature 0.9 + Top-P 0.3 = creative but restricted to only the most likely tokens.

✖ COMMON MISTAKE

Cranking both temperature and top-p high at once expecting more control usually just adds chaos.



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Structured Output **vs** JSON Mode

? WHAT

JSON Mode: guarantees the output is valid JSON but not any specific shape.

Structured Output: enforces a specific schema (defined fields, types, structure).

⚡ WHY THE CONFUSION

Both sound like "the model returns clean, parseable data."

🕒 WHEN IT MATTERS

Building production pipelines JSON mode alone can still give you wrong/missing fields; structured output enforces the exact contract your code expects.

💡 EXAMPLE

JSON mode might return `{"result": "yes"}` when you expected `{"status": "approved", "confidence": 0.9}`. Structured output forces that exact shape.

✖ COMMON MISTAKE

Relying on JSON mode alone for critical pipelines, then writing brittle code to handle unexpected shapes.



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Long-Term Memory **VS** Context Window

? WHAT

Context Window: the temporary working memory for one session.

Long-Term Memory: a separate persistent store the system retrieves from and feeds back into context.

✂ WHY THE CONFUSION

Both feel like "what the model remembers."

🕒 WHEN IT MATTERS

Designing personalized AI products long-term memory needs its own storage + retrieval system; it doesn't just "grow" the context window.

💡 EXAMPLE

A coding assistant remembering your preferred code style across projects = long-term memory feeding facts back into a fresh context window each time.

✖ COMMON MISTAKE

Assuming long context window = long-term memory. Without a memory architecture, everything is forgotten once the session ends.



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